

Stadium & Live Event Political Advertising Opportunities Analysis

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Introduction

This project analyzes political advertising opportunities in sports stadiums and live-event environments. Sports venues provide a unique advertising setting due to high attendance, extended dwell time, and strong regional audience engagement. Unlike traditional media, stadium advertising allows campaigns to repeatedly expose audiences to messaging throughout the duration of an event.

The purpose of this analysis is to evaluate how factors such as stadium size, advertising placement, engagement levels, and cost influence the effectiveness of political advertising. By using a data-driven approach, this project identifies which stadiums, locations, and advertising formats provide the most valuable opportunities for campaign messaging.

Dataset Description

The dataset used in this analysis includes multiple sports venues across different leagues, such as the NFL, MLB, NBA, NHL, and college football. Each observation represents a specific advertising opportunity within a stadium, including information about attendance, advertising type, placement, estimated audience reach, engagement score, and cost.

Key variables include stadium capacity, average attendance, advertising format (such as LED boards, ribbon boards, and concourse screens), estimated reach, engagement score, and cost of advertising. These variables allow for the evaluation of both exposure potential and cost efficiency across different venues and advertising formats.

Methodology

The analysis was conducted using Python in a Jupyter Notebook environment. The dataset was explored using Pandas, and visualizations were created using Matplotlib.

An important part of the analysis was the creation of an “Opportunity Score,” which combines estimated reach, engagement, and cost into a single metric. This score allows for the comparison of advertising opportunities across different stadiums and placements.

The Opportunity Score was calculated as:

$$\text{Opportunity Score} = (\text{Estimated Reach} \times \text{Engagement Score}) \div \text{Cost}$$

This metric helps identify which advertising opportunities provide the highest value relative to cost.

Exploratory Data Analysis

Before conducting deeper analysis, an initial exploration of the dataset was performed to understand its structure, distribution, and overall quality. The dataset contains 120 observations across multiple sports venues, with variables including stadium name, sport type, location, attendance, advertising format, placement, estimated reach, engagement score, and cost.

The dataset was complete with no missing values, ensuring consistency and reliability for analysis. Summary statistics revealed significant variation in key variables such as attendance, estimated reach, and cost, indicating that advertising opportunities differ widely across venues and placements.

Initial inspection also showed that college football stadiums tend to have the highest attendance levels, while smaller venues such as NBA and NHL arenas operate at lower capacity ranges. This variation is important, as it directly impacts estimated reach and overall advertising effectiveness.

Overall, the exploratory analysis confirmed that the dataset is suitable for evaluating advertising performance and provided a foundation for the more detailed analysis presented in the following sections.

Analysis and Findings

1. Venue Advertising Opportunity Score

[5]:

	Stadium	Sport	State	Capacity	Avg_Attendance	Ad_Type	Placement	Estimated_Reach	Engagement_Score	Cost_K	Opportunity_Score
0	Crypto.com Arena	NBA	CA	19000	17537	Scoreboard	Broadcast Integration	192354	8.12	74.99	20828.303507
1	Honda Center	NHL	CA	17100	16117	Concourse Screen	Upper Deck	83902	11.79	258.11	3832.492271
2	Madison Square Garden	NBA	NY	19800	13864	Concourse Screen	Broadcast Integration	154927	9.28	51.77	27771.345567
3	Wrigley Field	MLB	IL	41649	32550	Scoreboard	Field Level	177939	11.82	108.19	19440.234587
4	Michigan Stadium	College Football	MI	107600	98975	Scoreboard	Concourse	1003838	9.15	61.61	149084.851485

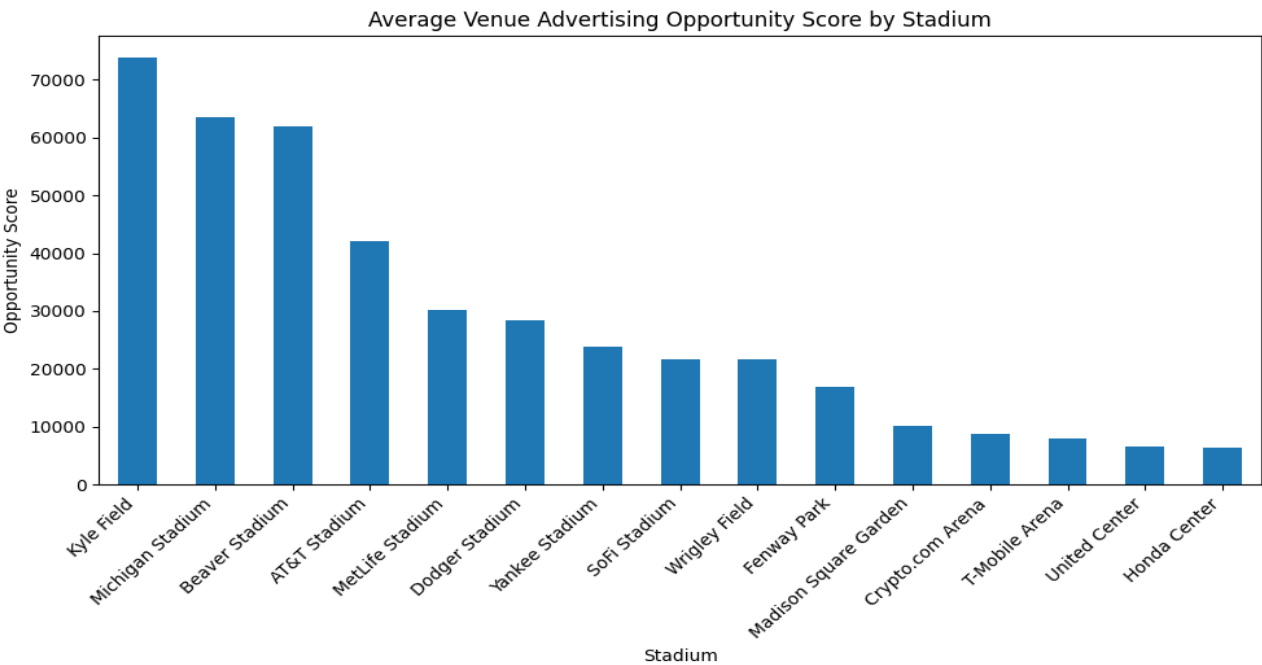


Figure 1. Average Opportunity Score by Stadium

This figure highlights the variation in advertising opportunities across different stadiums by combining reach, engagement, and cost into a single metric. Stadiums such as Kyle Field, Michigan Stadium, and Beaver Stadium show the highest opportunity scores, indicating strong performance driven by large audience sizes and relatively efficient advertising costs. In contrast, smaller venues or those with lower engagement levels produce significantly lower scores. This suggests that high-capacity stadiums, particularly in college football, offer the most valuable environments for political advertising campaigns due to their ability to deliver both scale and efficiency.

2. Estimated Reach by Stadium

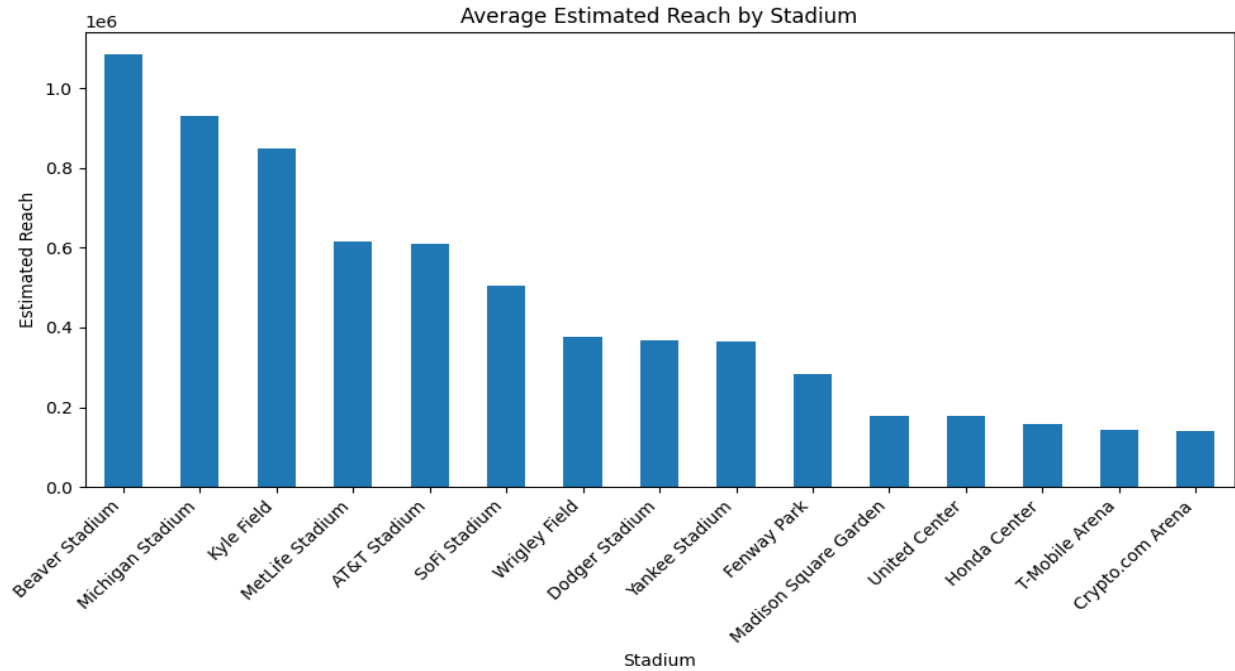


Figure 2. Average Estimated Reach by Stadium

This chart demonstrates a strong positive relationship between stadium attendance and estimated reach. As attendance increases, reach grows significantly, reinforcing the importance of venue size in advertising strategy. However, some variation exists, indicating that factors such as placement and engagement also influence reach. This confirms that while attendance is a key driver, it should be considered alongside other variables when evaluating advertising opportunities.

3. Engagement by Advertising Type

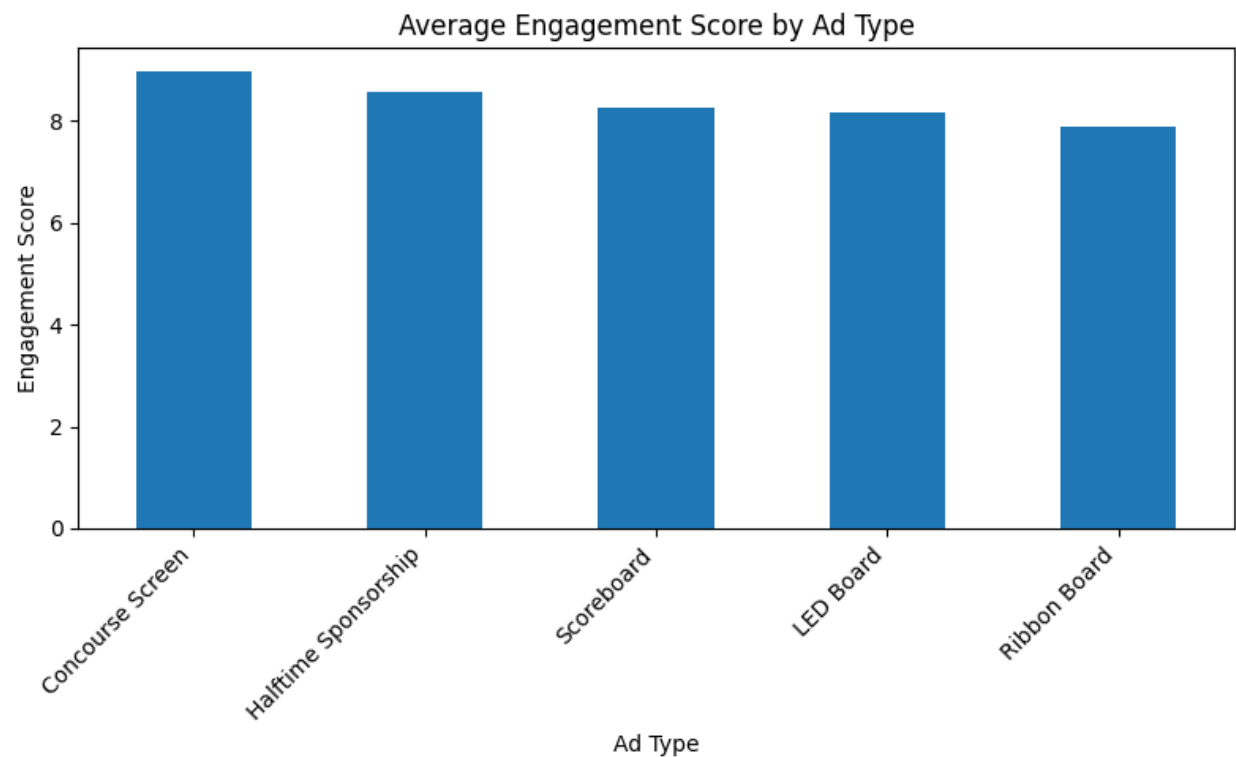


Figure 3. Average Engagement Score by Ad Type

This figure compares engagement levels across different advertising formats within stadiums. Concourse screens and halftime sponsorships show relatively higher engagement scores, suggesting that ads placed in high-traffic or high-attention areas are more likely to capture audience interest. In contrast, traditional placements such as ribbon boards show slightly lower engagement. These findings indicate that the format and placement of advertisements play a critical role in influencing audience interaction, not just visibility.

4. Advertising Opportunity by Placement

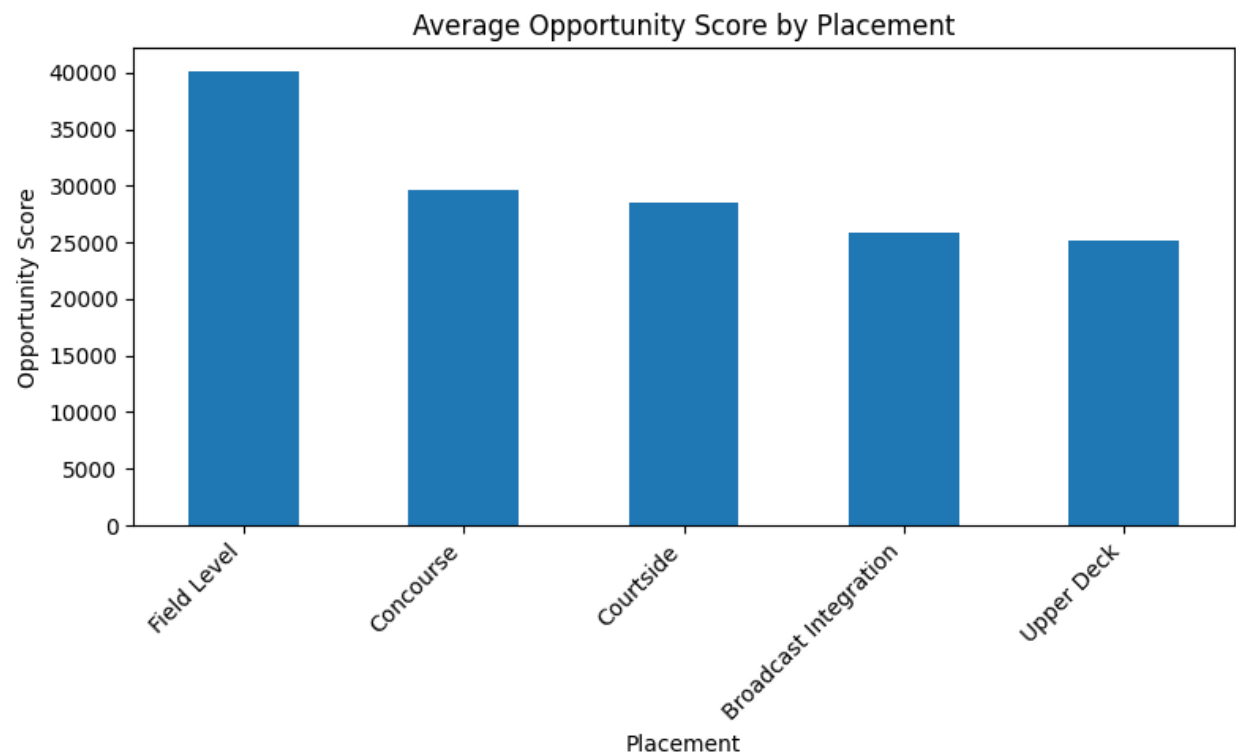


Figure 4. Opportunity Score by Placement

This chart evaluates how different in-stadium placements impact overall advertising effectiveness. Field-level placements generate the highest opportunity scores, followed by concourse and courtside placements. These locations likely benefit from closer proximity to fans and higher visibility during key moments of the event. Upper deck placements show lower performance, suggesting reduced engagement and visibility. This highlights the importance of premium placement when optimizing advertising strategy.

5. State-Level Advertising Opportunities

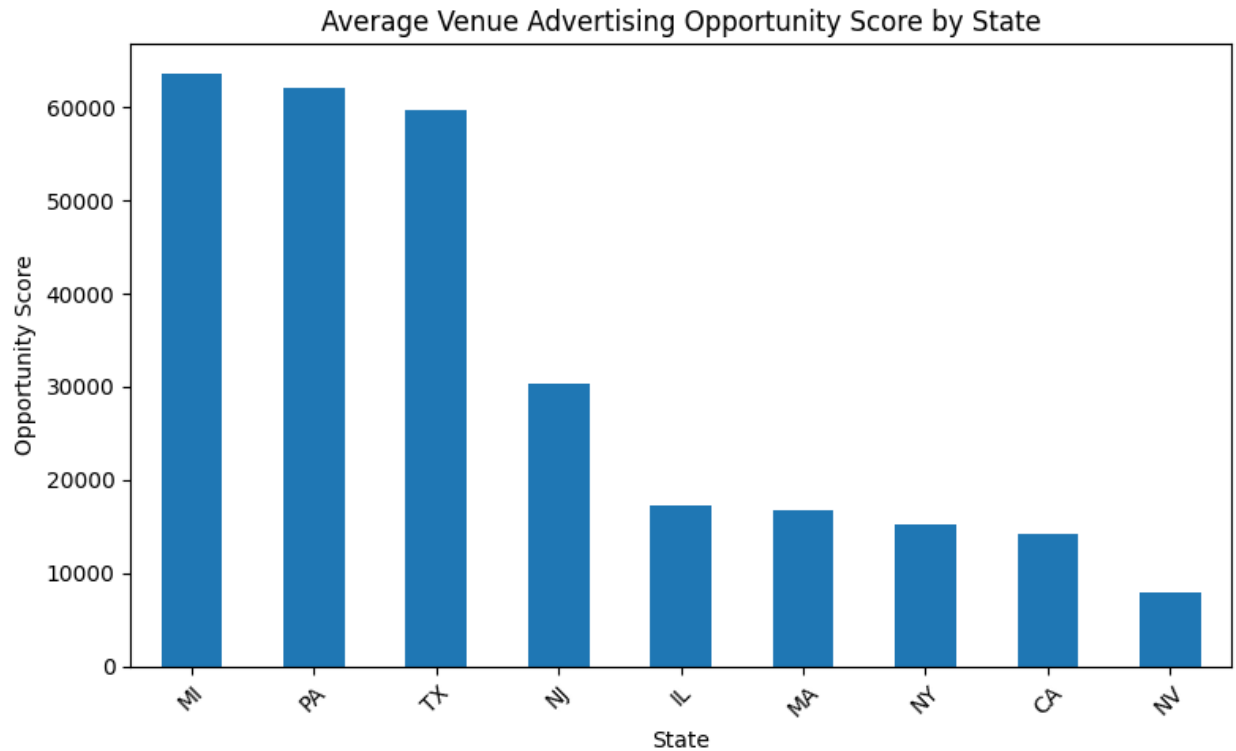


Figure 5. Average Opportunity Score by State

This figure shows differences in advertising opportunities across states. States such as Michigan, Pennsylvania, and Texas demonstrate the highest opportunity scores, largely due to the presence of large stadiums and strong attendance levels. These regions provide higher exposure and engagement potential, making them strategically valuable for targeted political advertising campaigns. In contrast, states with smaller venues or lower engagement levels offer reduced advertising impact.

6. Cost vs Estimated Reach

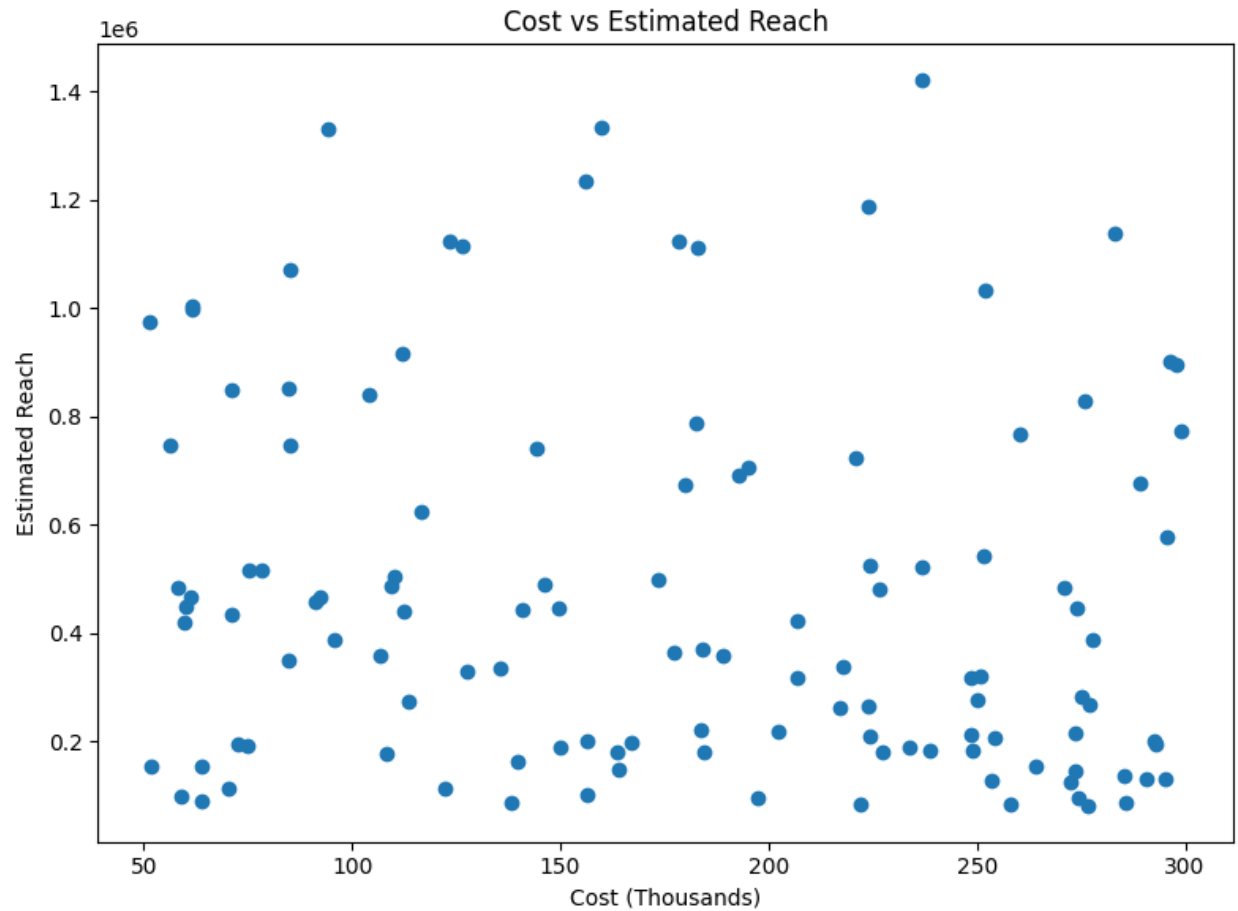


Figure 6. Cost vs Estimated Reach

This scatter plot explores the relationship between advertising cost and estimated reach. While there is a general trend where higher costs are associated with greater reach, the relationship is not perfectly linear. Several lower-cost data points still achieve relatively high reach, indicating the presence of cost-efficient advertising opportunities. This suggests that campaigns can optimize spending by identifying placements that deliver strong exposure without requiring maximum investment.

7. Attendance vs Estimated Reach

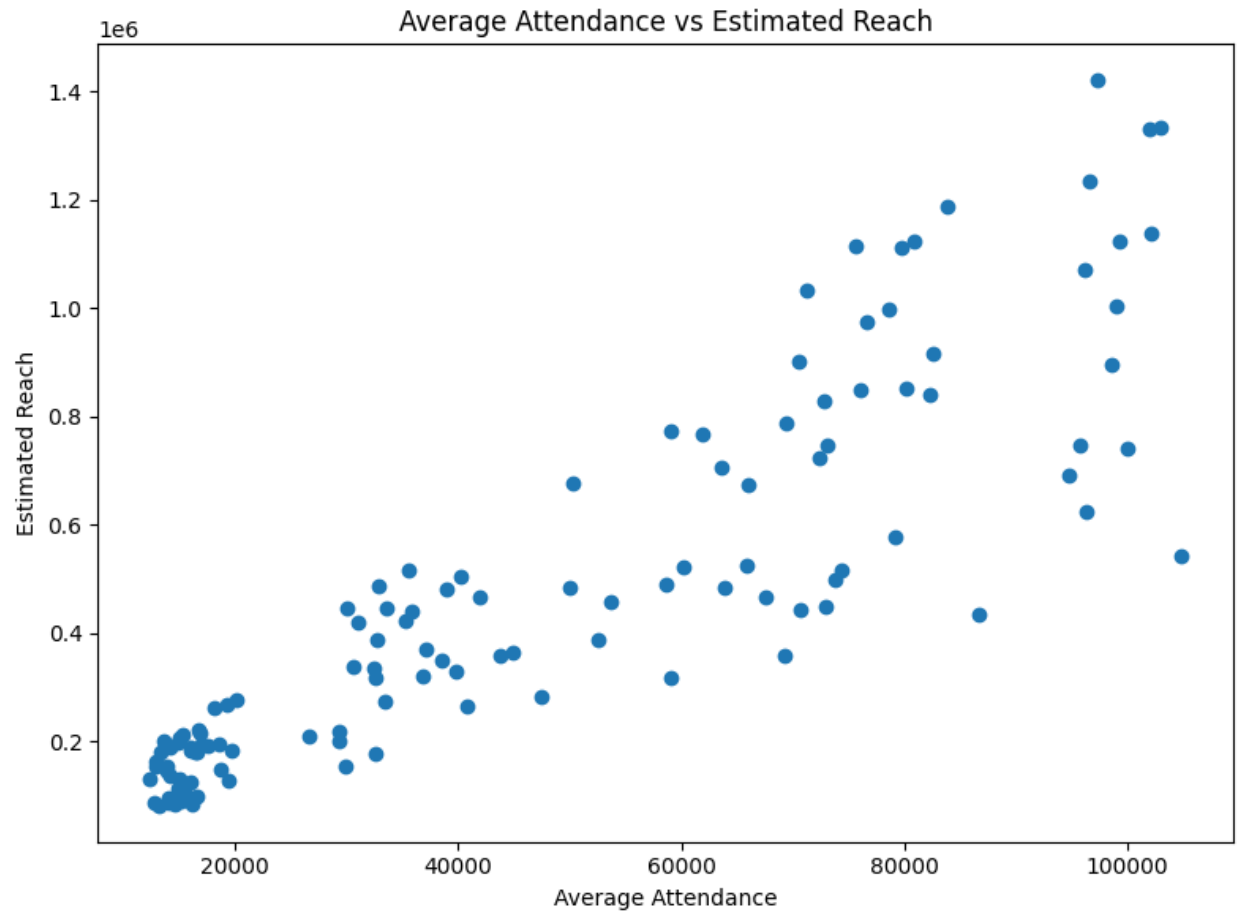


Figure 7. Attendance vs Estimated Reach

This chart demonstrates a strong positive relationship between stadium attendance and estimated reach. As attendance increases, reach grows significantly, reinforcing the importance of venue size in advertising strategy. However, some variation exists, indicating that factors such as placement and engagement also influence reach. This confirms that while attendance is a key driver, it should be considered alongside other variables when evaluating advertising opportunities.

8. Opportunity by Sport

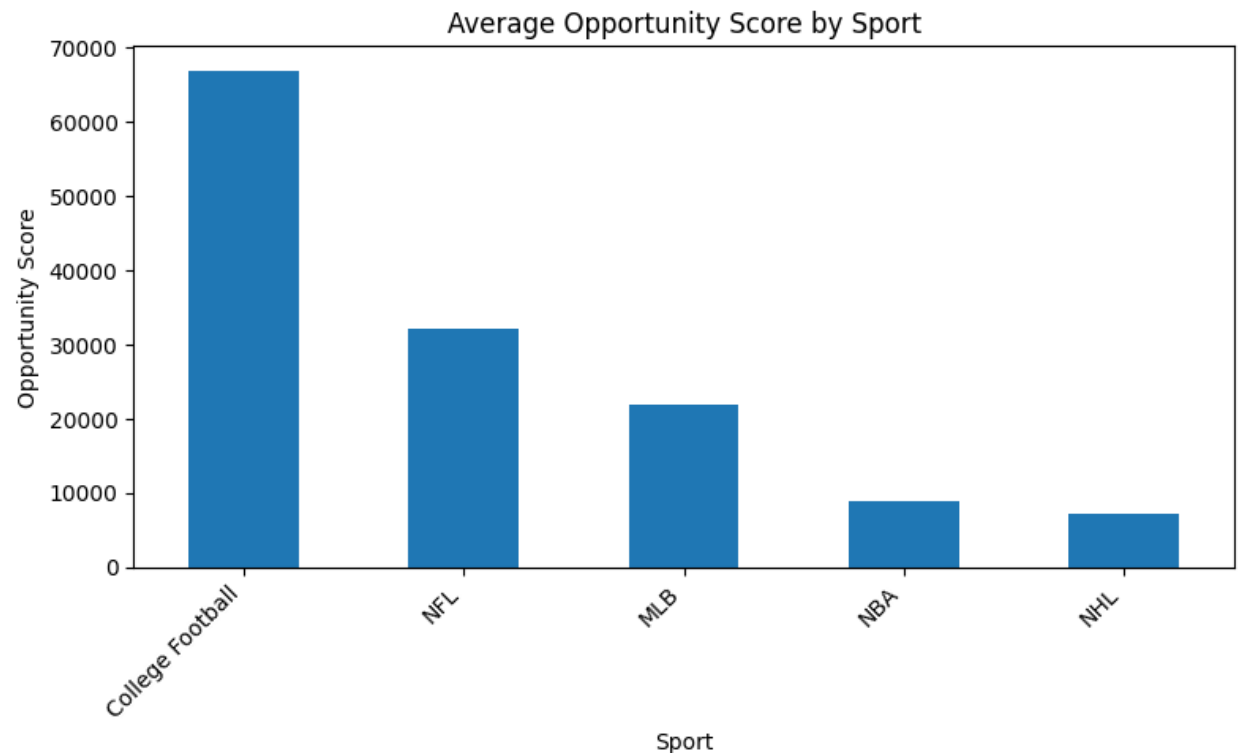


Figure 8. Average Opportunity Score by Sport

This figure compares advertising effectiveness across different sports. College football clearly stands out with the highest opportunity score, followed by the NFL and MLB. This is likely due to larger stadium capacities, longer event durations, and highly engaged fan bases. In contrast, sports such as the NBA and NHL show lower opportunity scores, reflecting smaller venues and different audience dynamics. This suggests that sport selection plays a significant role in maximizing advertising impact.

Key Insights

- Larger stadiums consistently provide higher advertising reach due to greater attendance.
- Certain advertising formats generate stronger engagement, increasing overall effectiveness.

- Placement within the stadium significantly impacts advertising performance.
- Some states offer more valuable advertising opportunities due to audience size and engagement.
- Cost efficiency varies widely, highlighting opportunities for optimizing advertising investment.

Strategic Recommendations

- Prioritize high-capacity stadiums with strong engagement scores.
- Focus on high-visibility placements such as LED boards and scoreboard displays.
- Allocate budget toward cost-efficient venues that offer strong reach relative to cost.
- Target states and regions with higher audience engagement and attendance levels.

Ethical Considerations

The use of data in political advertising, particularly within sports environments, raises several important ethical considerations. While data-driven strategies can improve targeting and efficiency, they must be applied responsibly to ensure fairness, transparency, and respect for individuals.

One key concern is privacy and data usage. Political advertising often relies on audience data such as location, demographics, and behavioral patterns. Any data used must be collected and processed in compliance with privacy regulations and with appropriate user consent. Unauthorized use of personal data or overly intrusive targeting practices can undermine public trust.

Another important issue is targeting and fairness. Highly targeted political advertising may disproportionately influence specific groups of voters, particularly those who are more susceptible to persuasion. This raises concerns about fairness and the potential for manipulation. Campaigns should ensure that messaging is not misleading and that it does not exploit vulnerable populations.

Additionally, there is the issue of transparency in political messaging. Viewers should be clearly informed when they are exposed to political advertisements, including who is

sponsoring the message. Lack of transparency can lead to misinformation and reduce accountability.

There are also ethical considerations related to the use of sports environments for political messaging. Sports venues are traditionally spaces for entertainment and community engagement, and the introduction of political advertising may affect the fan experience. Campaigns should carefully consider the balance between effective communication and maintaining a positive environment for audiences.

Finally, data scientists have a responsibility to ensure that their analyses are **accurate**, unbiased, and not misleading. The results of this project should be interpreted carefully, recognizing that the dataset is simulated and may not fully represent real-world conditions. Ethical data practice requires transparency about data limitations and avoidance of overgeneralization.

Conclusion

This analysis demonstrates that sports stadiums provide a highly effective environment for political advertising due to their ability to deliver sustained audience exposure and engagement. By combining attendance data, advertising placement, and cost analysis, this project identifies which venues offer the greatest value for campaign messaging.

The results show that successful advertising strategies require careful selection of stadiums, placement types, and geographic markets. Campaigns that leverage data-driven insights can optimize their advertising investments and improve overall effectiveness in reaching target audiences.